

Wave Dispersion and Transformation

Theory note — linear (Airy) wave theory: dispersion, shoaling and refraction

1. Linear wave theory and the dispersion relation

Linear (Airy) theory describes small-amplitude, sinusoidal waves on water of locally constant depth. For a wave of period T on still water of depth d , the angular frequency ω and the wavenumber k are linked by the **dispersion relation**:

$$\omega^2 = gk \tanh(kd), \quad \omega = 2\pi/T, \quad k = 2\pi/L,$$

where L is the wavelength and g the acceleration of gravity. In deep water this reduces to the deep-water length $L_0 = gT^2/2\pi$. The relation is implicit in k , so it is solved numerically; the calculator iterates $L = L_0 \cdot \tanh(2\pi d/L)$ to convergence.

2. Phase speed, group speed and wave regimes

The **phase speed** (celerity) c and the **group speed** c_g (the speed of energy transport) are

$$c = \omega/k = L/T, \quad c_g = n \cdot c, \quad n = \frac{1}{2}[1 + 2kd/\sinh(2kd)].$$

The relative depth d/L sets the regime:

Regime	d/L	n	$c \rightarrow$	$L \rightarrow$
Deep water	$> 1/2$	$1/2$	$gT/2\pi$	$L_0 = gT^2/2\pi$
Intermediate	$1/20 - 1/2$	$1/2 - 1$	(full relation)	(full relation)
Shallow water	$< 1/20$	1	$\sqrt{gd}$	$T\sqrt{gd}$

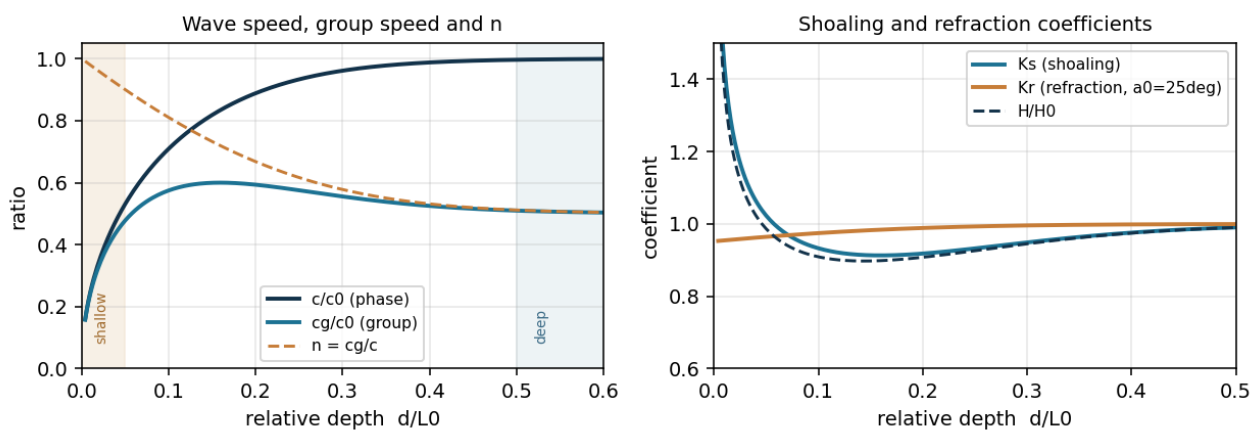


Figure 1. Left: phase speed, group speed and n versus relative depth (deep water at right, shallow at left). Right: shoaling and refraction coefficients and their product H/H_0 for a 25° approach.

3. Shoaling

The energy density (per unit surface area) and the wave power (per unit crest length) are $E = \frac{1}{2}\rho g A^2$ and $P = E \cdot c_g$, where $A = H/2$. For waves approaching normally over parallel contours the energy flux is conserved, $E \cdot c_g = \text{constant}$, so the height changes through the group speed, giving the **shoaling coefficient**

$$K_s = \sqrt{c_{g0}/c_g}, \quad \text{with } c_{g0} = c_0/2.$$

4. Refraction

Where the crest meets the depth contours at an angle, the change of c with depth turns the crest (refraction), governed by **Snell's law**:

$$\sin\alpha/c = \text{constant} \rightarrow \sin\alpha = (c/c_0) \cdot \sin\alpha_0, \quad K_r = \sqrt{(\cos\alpha_0/\cos\alpha)},$$

where α is the angle between the wave crest and the depth contour, and subscript 0 denotes deep water.

5. Combined transformation

For straight, parallel offshore contours the overall change in height from deep water to a local depth d is

$$H = H_0 \cdot K_s \cdot K_r.$$

6. Worked example

For $T = 10$ s, $d = 8$ m, $H_0 = 2$ m and $\alpha_0 = 25^\circ$: $L_0 = 156.1$ m and the solved $L = 83.8$ m, so $d/L = 0.095$ (intermediate). Then $c = 8.38$ m/s, $n = 0.898$, $c_g = 7.52$ m/s. $K_s = \sqrt{(7.807/7.52)} = 1.019$; the wave refracts to $\alpha = 13.1^\circ$ so $K_r = 0.965$. The transformed height is $H = 2 \times 1.019 \times 0.965 = 1.97$ m.

7. Notation

T, L — wave period and wavelength

ω, k — angular frequency ($2\pi/T$) and wavenumber ($2\pi/L$)

c, c_g — phase speed (celerity) and group speed

n — ratio c_g/c

H, A — wave height and amplitude ($A = H/2$)

α — wave angle to the shore-normal

K_s, K_r — shoaling and refraction coefficients

subscript 0 — deep-water value

8. References

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Companion to the Wave Dispersion & Transformation web calculator — M. N. Sahvelet, Troy Lab, Purdue University.